



10 CFR 50.73

CCN: 25-87

October 8, 2025

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Peach Bottom Atomic Power Station (PBAPS) Units 2 and 3
Subsequent Renewed Facility Operating License No. DPR-44, DPR-56
NRC Docket No. 50-277, 50-278

Subject: Licensee Event Report (LER) 2025-001-00 Invalid Actuation Due to Bus Failure

The subject report is being submitted in accordance with 50.73(a)(2)(iv)(A) for an invalid actuation of general containment isolation signals affecting containment isolation valves in more than one system.

There are no commitments contained in this letter. If you have any questions, please contact the Peach Bottom Regulatory Assurance Manager, Mr. Brian Woodard at (257) 533-7430.

Respectfully,

Ryan C. Stiltner
Plant Manager
Peach Bottom Atomic Power Station

Enclosure

cc: USNRC, Administrator, Region I
USNRC, Senior Resident Inspector
W. DeHaas, Commonwealth of Pennsylvania
S. Seaman, State of Maryland
B. Watkins, PSE&G, Financial Controls and Co-Owner Affairs



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Peach Bottom Atomic Power Station (PBAPS), Unit 2

☒ 050
☐ 052

2. Docket Number

00277

3. Page

1 OF 2

4. Title

Invalid Actuation Due to Bus Failure

5. Event Date

Month	Day	Year
08	20	2025

6. LER Number

Year	Sequential Number	Revision No.
2025	- 001 -	00

7. Report Date

Month	Day	Year
10	08	2025

8. Other Facilities Involved

Facility Name	☒ 050	Docket Number
PBAPS, Unit 3		00278
Facility Name	☐ 052	Docket Number

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Brian Woodard, Regulatory Assurance Manager

Phone Number (Include area code)

(267) 533-7430

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
E	EA	BU	P310	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On August 20th, 2025, an electrical fault in a metal enclosed bus duct resulted in the loss of startup switchgear (EIS: EA) that supplies power to administrative buildings and cooling towers. The electrical transient produced because of this event led to a partial invalid Primary Containment Isolation System (PCIS) (EIS: JM) isolation including isolation of the Reactor Water Cleanup System (RWCU) (EIS: CE) on both units. On Unit 2, this condition resulted in actuations of PCIS Group II and III outboard valves and isolation of the RWCU. On Unit 3, this condition resulted in actuations of PCIS Group II and III inboard valves and isolation of RWCU. Additionally, the 'A' and 'B' Cooling Tower Lift Pumps tripped.

Based on the as-found condition of the bus duct, the cause of failure is attributed to a compromised seal on the bus duct cover. The seal degraded over time, allowing water intrusion and ultimately led to a short circuit condition.

This condition is reportable under 50.73(a)(2)(iv)(A) as an invalid actuation of general containment isolation signals affecting containment isolation valves in more than one system. The isolation was caused by a loss of power. The actuation was invalid because the conditions for which isolation is required were not present.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME

Peach Bottom Atomic Power Station (PBAPS),
Unit 2

☒ **050**☐ **052****2. DOCKET NUMBER**

00277

3. LER NUMBER

YEAR	SEQUENTIAL NUMBER	REV NO.
2025	- 001	- 00

NARRATIVE**Plant Operating Conditions Before the Event**

Prior to the event, Peach Bottom Atomic Power Station, Units 2 and 3 were operating at approximately 100% power in MODE 1. There were no structures, systems or components (SSC) that were inoperable at the start of the event that contributed to the event, except as described below.

Event Description

On August 20th, 2025 at approximately 04:30am, an electrical fault in a metal enclosed bus duct resulted in the loss of startup switchgear (EIS: EA) that supplies power to administrative buildings and cooling towers. The electrical transient produced because of this event led to a partial invalid Primary Containment Isolation System (PCIS) (EIS: JM) isolation including isolation of the Reactor Water Cleanup System (RWCU) (EIS: CE) on both units. On Unit 2, this condition resulted in actuations of PCIS Group II and III outboard valves and isolation of RWCU. On Unit 3, this condition resulted in actuations of PCIS Group II and III inboard valves and isolation of RWCU. Additionally, the 'A' and 'B' Cooling Tower Lift Pumps tripped.

During the initial inspection of the bus duct, multiple degraded connections were observed. These deteriorations resulted from water intrusion through a degraded seal, leading to corrosion. The degraded seal allowed moisture to enter the bus duct, accelerating the corrosion process and adversely affecting the integrity of the electrical connections. The bus duct heater, which is designed to minimize condensation in the bus assembly, was found to be non-functional.

This condition is reportable under 50.73(a)(2)(iv)(A) as an invalid actuation of general containment isolation signals affecting containment isolation valves in more than one system. The isolation was caused by a loss of power. The actuation was invalid because the conditions for which isolation is required were not present.

Safety Consequences

There were no challenges to plant safety as a result of this event. Both units responded as designed to the electrical transient, and isolations were reset without error.

Corrective Actions

Corrective actions for this condition include installation of new bus bar and bus duct and establishing a routine inspection of this portion of the bus.

Previous Similar Events

Peach Bottom experienced invalid actuations due to an emergency bus breaker relay failure in 2019, ref. LER 2019-001-00.